# HoloAssist – “Futurino UR3eno”

**Client:** Augustine Le

**Coach:** Tan Huynh

**Revised:** 16/4/26

**Group Name:** Thu_9_12_00

**Team Members:**

| Full Name | Student ID | Expertise |
|---|---|---|
| John Chen | 24837332 | Perception pipeline, RGB-D camera integration, object detection & classification, ROS 2 publish, coordinate frame calibration |
| Sebastian Baudille | 24804940 | XR scene & visualisation, Unity 6 environment, virtual object overlay system, UI/UX, coordinate frame alignment |
| Oliver Lau | 24770051 | Autonomous sorting, MoveIt 2 trajectory planning, pick-and-place execution, gripper integration, mode switching |
| Nicholas Sabatta | 24796955 | Teleoperation & interaction, XR controller input, RMRC/Jacobian control, gripper control, e-stop dashboard, session logging |

# Overview

## 1.1 Project Description

HoloAssist is an XR-assisted robotic sorting system operating in a shared physical workspace. A UR3e collaborative arm, controlled via a Meta Quest headset, sorts physical objects into bins within a defined workspace. An Intel RealSense RGB-D depth camera detects objects as they enter the workspace, identifies them by colour or computer vision, and tracks their position until the robot manipulates them. The camera is intended to be mounted in a stable fixed configuration using a tripod or bracket to improve repeatability of perception, calibration, and workspace monitoring. The system is designed for human-robot collaboration, where a human operator in a headset may control the robot while another person may also be present in the real environment, requiring layered safety precautions including velocity limits, collision-aware behaviour, safe stop functionality, and emergency stop access during runtime.

The system operates in two modes:

- **Teleoperation Mode — a human operator wearing the Quest 3 headset controls the robot arm and gripper to pick up objects and place them in the correct bin. The operator sees the workspace in mixed reality with XR overlays showing object identity, bin assignments, and robot state. Teleoperation is supported by safety mechanisms including joint speed limits, velocity scaling, safe stop behaviour, and emergency stop access through the Steam Deck dashboard.**
- **Autonomous Mode — the robot autonomously sorts objects using MoveIt 2 planned trajectories without human input. Object-bin assignments come from a predefined mapping.**

## 1.2 XR Integration

Real-world objects are mapped to different virtual objects in the XR scene. For example, a physical tomato on the workspace appears as a bomb in the Quest 3 headset that the operator must "defuse" by sorting it into the correct bin. This creates an engaging, gamified operator experience while the underlying robotic task remains a practical sorting pipeline.

### 1.3 Stretch Goal — Human vs CPU

Teleoperation session data (object poses, grasp strategies, sort times) can be used to enable a "race mode" where a teleoperator competes against an overlay of the autonomous agent performing the same task — primarily for demonstration purposes.

## 2. System Architecture

### 2.1 High-Level Data Flow

The system is structured around four integrated subsystems communicating via ROS 2:

- Object enters workspace — the RealSense RGB-D camera detects, identifies, and publishes its pose to ROS 2.
- Pose broadcast — the object pose is transformed into the robot coordinate frame via QR code calibration and forwarded to Unity and RViz.
- XR rendering — Unity receives the pose and renders the corresponding virtual object at the correct location in the operator's mixed reality view.
- Sorting — in teleoperation mode a human operator wearing the headset controls the robot to approach, grasp, and sort the object; in autonomous mode MoveIt 2 plans and executes the pick-and-place trajectory.
- Workspace monitoring — the fixed RGB-D camera continuously observes the workspace to support object tracking, bin verification, and safety-aware monitoring of the task area.
- Bin verification — the camera confirms whether the object was placed in the correct bin.
- Runtime safety response — if communication, control, or operator conditions require intervention, the system supports safe stop behaviour and emergency stop access via the operator dashboard.

### 2.2 Subsystems

#### Subsystem 1 — Perception and Mapping (Lead: John Chen)

Object detection, identification, spatial tracking, and workspace monitoring using the Intel RealSense RGB-D camera.

- ☐ Configure and calibrate the RealSense (intrinsics, extrinsics, camera-to-robot transform via QR codes)
- ☐ Detect objects entering the workspace via depth segmentation and background subtraction
- ☐ Identify objects by colour, shape, or computer vision classification
- ☐ Publish object poses to ROS 2 in the robot coordinate frame
- ☐ Track objects through the workspace until manipulated
- ☐ Optionally verify correct bin placement after sorting
- ☐ Support workspace monitoring for safety-aware operation, including stable observation of the robot task area and optional verification of object placement or intrusion into defined regions

## Subsystem 2 — Autonomous Sorting (Lead: Oliver Lau)

Autonomous pick-and-place using MoveIt 2 trajectory planning.

- ☐ Configure MoveIt 2 for the UR3e (collision geometry, planning groups, end-effector)
- ☐ Plan collision-free trajectories from current pose to detected object pose
- ☐ Execute pick-and-place sequences: approach → grip → lift → move to bin → release
- ☐ Integrate gripper open/close commands into the motion pipeline
- ☐ Handle object-to-bin assignment logic (predefined mapping)
- ☐ Support mode switching between teleoperation and autonomous sorting

## Subsystem 3 — Teleoperation and Interaction (Lead: Nicholas Sabatta)

XR-based teleoperation — the human operator controls the robot arm and gripper through the Quest headset while monitoring system state through the XR view and supporting dashboard interfaces. This subsystem is responsible for interactive control, operator input handling, session logging, and runtime safety intervention.

- ☐ Three teleoperation modes (existing, fully functional):
  - ☐ RMRC — Jacobian-based Cartesian velocity control with translate/rotate sub-modes
  - ☐ Direct Joint — individual joint jogging
  - ☐ Hand Guide — controller position tracking with Jacobian IK
- ☐ New: Gripper control — trigger or button to open/close the OnRobot gripper via ROS
- ☐ E-stop dashboard on Steam Deck (existing — fully functional)

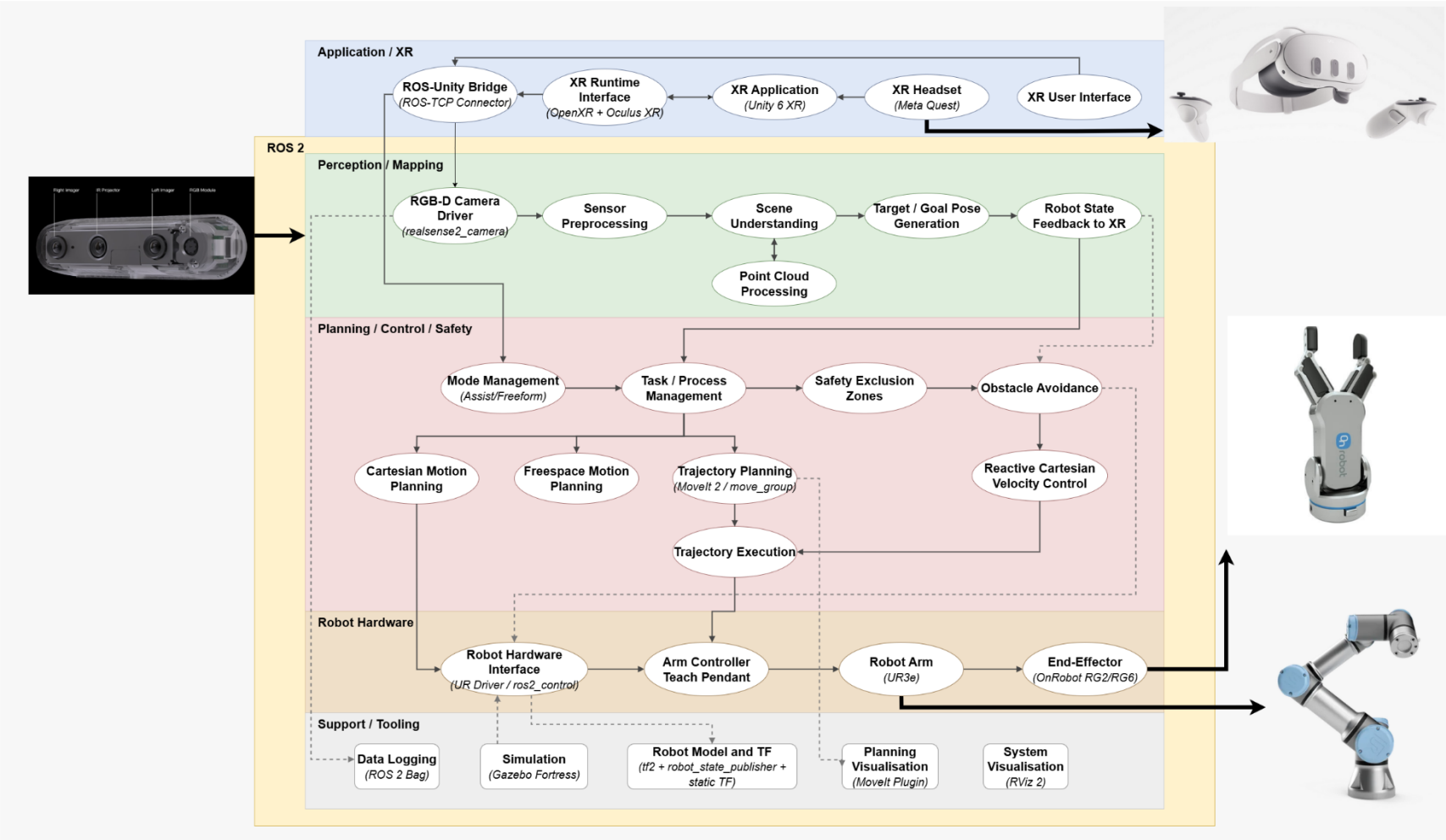
- [ ] Session logging and telemetry (existing — mode tracking, durations, events)
- [ ] Safety features: joint speed limits, velocity scaling, safe stop behaviour, e-stop access on Steam Deck, and support for operation in a shared human workspace

## Subsystem 4 — XR Scene and Visualisation (Lead: Sebastian Baudille)

Unity scene rendering, XR environment, and the virtual object overlay system.

- [ ] Design and build the XR scene environment in Unity 6
- [ ] Implement the object mapping system: real object type → virtual XR object (e.g. tomato → bomb)
- [ ] Render virtual objects at the correct pose in the MR passthrough view
- [ ] Visualise bins/targets in XR (labels, highlights, placement guides)
- [ ] Coordinate frame alignment between Unity scene and real workspace via QR code anchoring
- [ ] UI/UX for the operator: bin assignments, sorting progress, task status overlays
- [ ] Visual feedback on sort correctness (success/failure indicators)

## 1.2 High-level System Diagram



# Project Deliverables

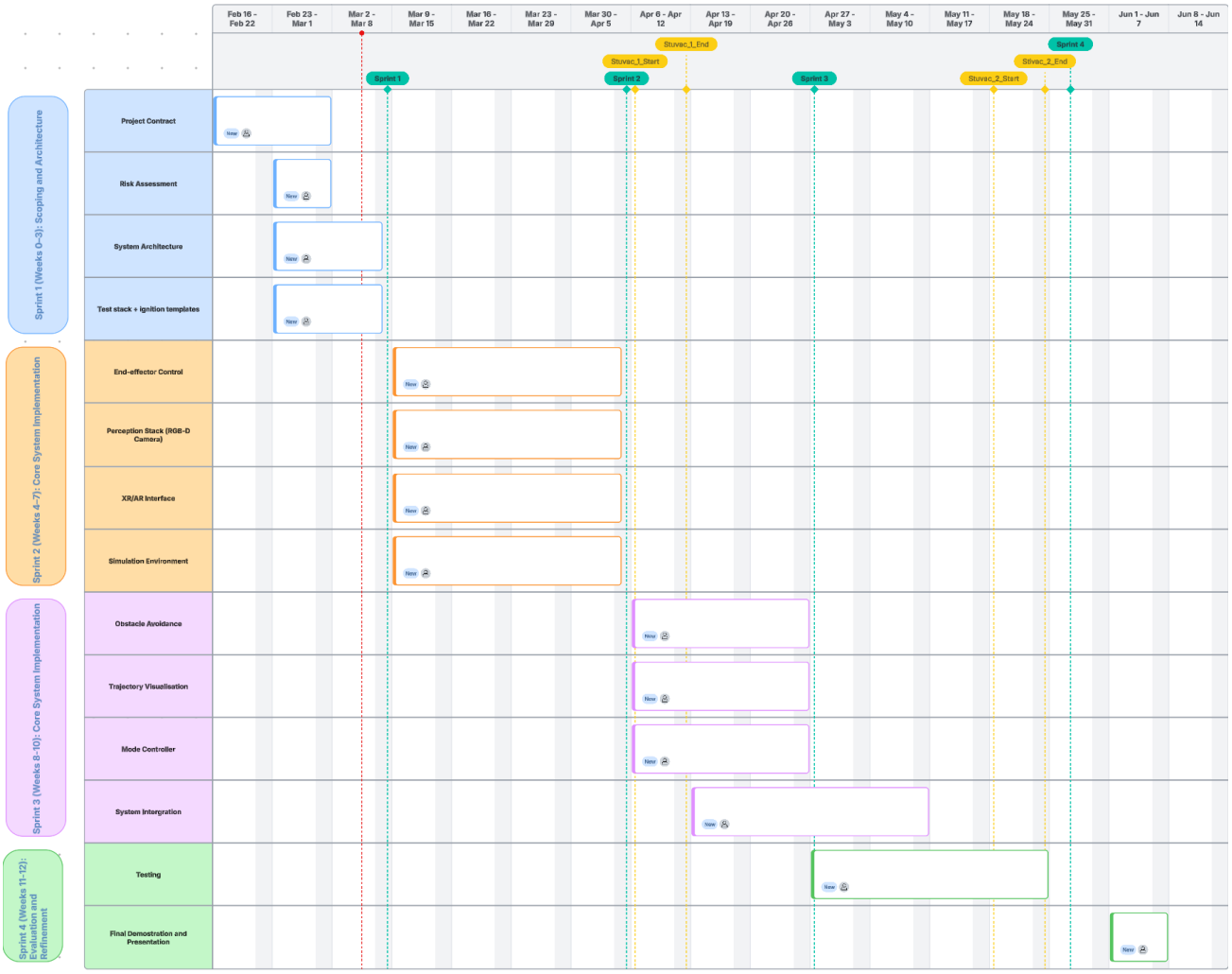
| ID | Deliverable | Description |
|---|---|---|
| D1 | End-Effector Control | Reactive Cartesian velocity control (RMRC) enabling safe teleoperation of the UR3e, including joint speed limits, velocity scaling, and safe stop on communication loss. |
| D2 | Perception Pipeline | Object detection, identification (colour/CV), and pose estimation using the Intel RealSense RGB-D camera, publishing to ROS 2 in the robot coordinate frame. |
| D3 | Autonomous Sorting | MoveIt 2–based pick-and-place pipeline that autonomously sorts detected objects into assigned bins without human input. |
| D4 | XR/AR Interface | Meta Quest 3 mixed reality interface displaying robot state, virtual object overlays (real→virtual mapping), bin assignments, and task status. |
| D5 | Mode Management | Framework supporting teleoperation mode, autonomous sorting mode, and mode switching; including e-stop dashboard on Steam Deck, safe stop handling, and runtime operator intervention. |
| D6 | Demonstration | Integrated system demonstration performing a validated sorting task in both teleop and autonomous modes. |
| D7 | Testing | Quantitative evaluation measuring sort accuracy, task completion time, safety interventions, and teleop vs autonomous performance comparison. |
| D8 | Documentation | Complete system architecture document, subsystem interface definitions, and risk assessment aligned with studio requirements. |

## 1.3 Key Milestones

### 1.3.1 Sprint 1 (Weeks 0–3): Scoping and Architecture

| Phase | Key Tasks |
|---|---|
| Phase 1 — Foundation (largely complete) | UR3e digital twin in Unity; XR teleoperation (RMRC, Direct Joint, Hand Guide); e-stop dashboard on Steam Deck; session logging and telemetry; headset streaming to dashboard. Remaining: OnRobot gripper driver in ROS; gripper control from XR controller. |
| Phase 2 — Perception Pipeline | RealSense configured and publishing in ROS; camera-to-robot calibration via QR codes; object detection and classification (colour/CV); object pose published in robot coordinate frame. |
| Phase 3 — Sorting Task | Object poses received in Unity and rendered as XR overlays; teleop sorting (operator picks and places into bins); autonomous sorting (MoveIt 2 pick-and-place); bin verification via depth camera. |
| Phase 4 — Integration & Polish | Mode switching between teleop and autonomous; XR object mapping (real→virtual, e.g. tomato→bomb); gamification overlay — Human vs CPU race (stretch goal); WiFi dropout auto e-stop (stretch goal); integrated shared-workspace demonstration with validated safety behaviour. |

## Time-Action Plan (Gantt Chart)



## 6.2 Software

| ID | Software | Purpose |
|---|---|---|
| S1 | Ubuntu 22.04 + ROS 2 Humble | Core OS and robotics middleware |
| S2 | MoveIt 2 | Autonomous trajectory planning and execution for pick-and-place |
| S3 | UR ROS2 Driver | Interface between UR3e controller and ROS 2 |
| S4 | Unity 6 | XR scene development, virtual object rendering, operator UI |
| S5 | ROS–Unity Bridge | Bidirectional communication between ROS 2 and Unity (ros_tcp_endpoint) |
| S6 | RealSense ROS2 Packages | RGB-D data acquisition, point cloud generation |
| S7 | OpenCV / PCL | Image processing, object classification, point cloud filtering |
| S8 | Gazebo Fortress | Simulation for development and testing without hardware |
| S9 | rosbag2 | Data logging for session replay, testing, and training data capture |
| S10 | Git / GitHub | Version control and collaborative development |

# 1.4 Learning Goals

### 1.4.1 Team Learning Goals

As a team, we aim to:

- Apply systems engineering principles to design, integrate, and validate a multi-subsystem robotics solution.
- Develop and evaluate a robotics path planning and control framework within a real-world collaborative context.
- Strengthen interdisciplinary collaboration through clearly defined subsystem ownership and interface agreements.
- Improve technical communication through sprint reviews, documentation, and reflective critique.
- Develop resilience in debugging and integration of complex robotic systems.

### 1.4.2 Individual Learning Goals

Each team member aims to deepen expertise within their subsystem while strengthening integration awareness:

- Develop advanced understanding of reactive motion control and constraint-based robotics.
- Gain practical experience integrating perception data into real-time control frameworks.
- Improve XR-based interface design for robotics applications.
- Strengthen hardware–software integration and safety-critical system thinking.
- Enhance professional capabilities in teamwork, reflection, and iterative design improvement.

# Resources

The successful completion of this project is incumbent on access to robotics hardware, perception systems, XR equipment, fabrication facilities, and supporting software frameworks. The required resources are outlined below.

## 1.5 Hardware and Software (Laboratory Provided)

| ID | Hardware | Quantity | Description |
|---|---|---|---|
| H1 | UR3e Collaborative Robot | 1 | 6-DOF Universal Robots UR3e collaborative manipulator used as the primary robotic platform for teleoperation, velocity control implementation, and collaborative task execution. |
| H2 | Mobile Robot Workstation | 1 | Trolley-mounted UR3e setup, including base frame and workspace surface, enabling flexible repositioning within the lab environment while maintaining safety compliance. |
| H3 | UR3e Teach Pendant | 1 | Official Universal Robots teach pendant used for robot configuration, safety setup, emergency stops, manual jogging, and recovery operations during development and testing. |
| H4 | Intel RealSense RGB-D Camera | 1 | RGB-D depth camera used for real-time point cloud generation, workspace sensing, object detection, and safety-aware monitoring of the robot task area. |
| H5 | Camera Mount / Tripod | 1 | Adjustable tripod or mounting bracket for stable positioning of the RGB-D camera in either eye-in-hand or eye-to-hand configuration, supporting repeatable calibration, consistent perception coverage, and reliable workspace monitoring. |
| H6 | Parallel Gripper | 1 | Two-finger end-effector for executing simple collaborative manipulation tasks (e.g., object alignment, placement, or handover demonstrations). |
| H7 | Workstation PC | 1 | High-performance desktop workstation running ROS 2, perception processing, control stack, and XR communication bridge. |
| H8 | Meta Quest 2 Headset | 1 | XR headset used for augmented reality visualisation of robot pose, point clouds, trajectories, safety boundaries, and headset-based teleoperation interface testing. |
| H9 | USB-C to Ethernet Adapter | 1 | An adapter that enables stable wired network connectivity for the Meta Quest headset, reducing wireless latency and improving communication reliability. |
| H10 | Ethernet Cables | 2–3 | CAT6 Ethernet cables for direct communication between workstation, UR3e controller, networking hardware, and XR device. |

| ID | Software | Description |
|---|---|---|
| S1 | Ubuntu 22.04 LTS & ROS 2 Humble | Core operating system and robotics middleware used for communication between perception, control, and XR subsystems. |
| S2 | MoveIt 2 | Motion planning framework used for robot modelling, kinematics, and optional trajectory validation. |
| S3 | Universal Robots ROS2 Driver | Interface enabling communication between the UR3e robot controller and the ROS 2 environment. |
| S4 | RViz2 | 3D visualisation tool used for debugging robot states, transforms, trajectories, and point cloud data. |
| S5 | RealSense ROS2 Packages | Drivers for acquiring RGB-D data and generating point clouds from the Intel RealSense camera. |
| S6 | Gazebo | A simulation tool to support the visualisation and testing of the system virtually without hardware. |
| S7 | Unity 6 | A game engine is used to develop the XR interface and visualise robot state and workspace information. |
| S8 | Unity XR Interaction Toolkit | Toolkit used within Unity to handle headset tracking and controller input for XR interaction. |
| S9 | ROS–Unity Communication Bridge | Middleware enabling communication between ROS 2 nodes and the Unity XR environment. |
| S10 | tf2 | ROS transform library used to manage coordinate frames between the robot, camera, and XR reference systems. |
| S11 | PCL (Point Cloud Library) | Used for filtering and processing real-time point cloud data. |
| S12 | OpenCV | Image processing library for filtering and validating RGB-D image data. |
| S13 | rosbag2 | A logging tool that supports the record of system data for testing, debugging, and evaluation. |
| S14 | Git and GitHub | Version control platform used for collaborative development, code management, and subsystem integration. |

## 1.6 Modifications to Laboratory Hardware

| ID | Modification | Description | Purpose |
|---|---|---|---|
| M1 | Custom Tool Holder / Gripper Adapter | Design and fabrication of a task-specific attachment compatible with the UR3e parallel gripper. | Enables execution of the selected demonstration task and mimics a simplified industry-relevant application. |
| M2 | RGB-D Camera Mounting Bracket | Design and fabrication of a secure mounting solution for the Intel RealSense camera. Mount location may be on the robot base, workspace frame, or end-effector, depending on chosen configuration (eye-in-hand or eye-to-hand). | Ensures stable depth sensing, repeatable calibration, accurate spatial alignment, and reliable workspace monitoring for perception and safety boundary generation. |

All modifications will comply with laboratory safety procedures and be reviewed with the lab manager before being added to the existing lab hardware.

## 1.7 Hardware and Software (Team-Sourced)

| ID | Item | Type | Description |
|---|---|---|---|
| TS1 | Meta Quest 2 | Hardware | XR headset used for augmented reality visualisation and teleoperation interface testing. |
| TS2 | Meta Quest 3 | Hardware | An optional upgraded XR headset is used to improve tracking stability and display performance during testing and demonstrations. |
| TS3 | Unity Hub & Unity 6 | Software | Development environment for building and managing the XR interface and spatial rendering pipeline. |
| TS4 | ROS–Unity Communication Bridge | Software | Middleware enabling bidirectional communication between ROS 2 nodes and the Unity XR environment (e.g. ROS-TCP-Connector or equivalent). |
| TS5 | Dedicated Networking Hardware (Optional) | Hardware | An optional router or wired networking setup to improve communication stability and reduce latency between the XR device and the control workstation. |

# 7. System Evaluation

| Level | Criteria |
|---|---|
| Pass (MVP) | XR headset displays stable robot model and end-effector pose. Operator can teleoperate robot using reactive velocity control. System stops safely on communication loss or operator intervention. Depth camera publishes to desktop UI and supports workspace monitoring. Operator can observe pose markers/transforms in XR. |
| Credit | Autonomous mode operational — MoveIt 2 plans and executes basic pick-and-place. Teleop sorting mode achievable in repeated trials via XR input. Gripper open/close integrated from XR controller. |
| Distinction | XR virtual object mapping operational (e.g. tomato → bomb). Mode switching between teleop and autonomous validated. Demonstration task completed consistently under defined time threshold. Passthrough / spatial marker visualisation working. |
| High Distinction | Human vs CPU race mode overlay functional. Quantitative comparison between teleop and autonomous modes (sort time, accuracy). Mixed reality UI visualises and debugs depth camera feed. WiFi dropout auto e-stop implemented. Demonstration shows safe operation in a shared workspace with clear intervention behaviour. |

# 8. Subsystem Evaluation

## Subsystem 1 — Perception (Lead: John Chen)

Object detection, identification, and spatial tracking using the Intel RealSense depth camera. Provides the perception pipeline that feeds both the teleoperation and autonomous sorting modes.

| Grade | Summary | Criteria |
|---|---|---|
| P | Camera publishing in ROS | • RealSense depth detects the workbench plane.<br>• Two AprilTags on the front corners define a stable workspace frame relative to the camera.<br>• The system publishes the workspace frame for other subsystems.<br>• The plane, frame, and raw pointcloud can be viewed in RViz or Foxglove. |
| C | Object detection and segmentation | • The system detects and tracks objects in the workspace.<br>• Depth is used to segment objects within the workspace region.<br>• Objects are identified by colour.<br>• An approximate object pose is published for Unity or robot planning. |
| D | Classification and pose in robot frame | • The pointcloud is filtered to remove irrelevant regions outside the bench workspace.<br>• The bench surface is suppressed while points above it are kept for tracking.<br>• Foreground objects are tracked over time, not just detected frame by frame.<br>• The pipeline maintains a usable and repeatable publish rate for real-time integration.<br>• Debug topics and visualisation outputs remain stable in RViz and Foxglove. |
| HD | Real-time tracking and bin verification | • Workspace perception remains stable under normal scene changes such as lighting variation, small occlusions, and nearby human activity.<br>• The system can distinguish meaningful intrusions, such as a human hand entering the workspace, from background points.<br>• Perception outputs are stable enough for safety-aware workspace monitoring.<br>• The subsystem shows reliable frame estimation, filtering, tracking, and pose publishing across repeated trials. |

## Subsystem 2 — Autonomous Sorting (Lead: Oliver Lau)

Autonomous pick-and-place sorting using MoveIt 2 trajectory planning. The robot sorts detected objects into the correct bins without human input.

| Grade | Summary | Criteria |
|---|---|---|
| P | MoveIt 2 coordinate planning | • Trajectory planning and execution demonstrated in RViz simulation<br>• MoveIt 2 trajectory planning and movement to specified locations |
| C | Collision detection/avoidance + gripper driver | • Collision boundaries (at minimum, the workspace table) prevent unsafe trajectories<br>• OnRobot gripper ROS driver functional — open/close commands work<br>• Gripper opens/closes as part of a scripted sequence (approach → grip → lift) |
| D | pick-and-place of object anywhere in the environment and place it in a set spot | • Mimics the object pose subscription to specify object locations<br>• Full pick-and-place cycle: approach → grip → lift → move to bin → release<br>• Object-to-bin mapping is defined<br>• Sequence is repeatable across multiple objects and trials |
| HD | Multi-object pick-and-place and sort out the objects | • Multiple objects sorted in a single autonomous session without manual resets<br>• Collision avoidance includes bins, workspace boundaries, and known obstacles<br>• Sorting metrics logged: objects sorted, success/failure per object, cycle time per sort<br>• Mode switching between teleop and autonomous is functional |

## Subsystem 3 — Teleoperation and Interaction (Lead: Nicholas Sabatta)

XR-based teleoperation of the UR3e arm and OnRobot gripper via Meta Quest 3. Includes control modes, operator dashboard, session logging, and gripper integration.

| Grade | Summary | Criteria / Evidence |
|---|---|---|

| | | |
|---|---|---|
| P | Stable digital twin on Quest 3 | ☐ JointStateSubscriber.cs subscribes to /joint_states and drives Unity joint transforms<br>☐ Digital twin verified on Quest 3 — joints match real robot direction and axes<br>☐ Purely kinematic (no physics simulation) — stable under all conditions |
| C | RMRC + Direct Joint + e-stop dashboard + Gripper | ☐ RobotController.cs with RMRC (Jacobian-based Cartesian velocity, translate/rotate sub-modes) and Direct Joint mode<br>☐ Both modes publish to /forward_velocity_controller/commands at 50 Hz<br>☐ UR3eKinematics.cs provides forward kinematics, geometric Jacobian, and DLS pseudoinverse<br>☐ Dashboard (main.py + ros_interface.py) with e-stop: burst zero-velocity publish, controller deactivation, 5-second hold resume, cooldown<br>☐ OnRobot gripper open/close triggered from XR controller during teleoperation, enabling full pick-and-place workflow |
| D | Three modes + headset streaming dashboard | ☐ Three control modes cycled via Menu button: RMRC → Direct Joint → Hand Guide<br>☐ RMRC has two sub-modes (Translate/Rotate) toggled via X button<br>☐ HeadsetStreamPublisher.cs renders VR camera view → JPEG → /headset/image_compressed at 15 FPS<br>☐ Dashboard tabs: STATUS, HEADSET (live XR view), STATS (session bar + joint velocity graph), LATENCY, SESSION<br>☐ Launch scripts (launch.sh, dashboard.sh) for one-command startup |
| HD | Hand Guide mode + session logging | ☐ Hand Guide mode: hold right grip to engage — controller 3D position mapped to end-effector via proportional control + Jacobian IK, gain 2, speed cap 0.15 m/s<br>☐ SessionLogger.cs: tracks mode switches, per-mode durations, session time — publishes JSON to /session/status (2 Hz) and saves on quit<br>☐ Dashboard saves session log to ~/holoassist_sessions/ on shutdown (events, e-stop count, Unity session info) |

## Subsystem 4 — XR Scene and Visualisation (Lead: Sebastian Baudille)

Unity XR environment design, virtual object overlay system, and operator-facing visualisation. Responsible for rendering the mixed reality scene on the Quest 3, including the real-to-virtual object mapping that creates the gamified sorting experience.

**Current:**

| Subsystem 4 – **XR and Scene Visualisation** | **Lead by: Sebastian** |
|---|---|
| P | Operator can observe pose markers and robot transforms in XR or desktop visualisation. |
| C | System visualisation supports repeated demonstration trials by clearly presenting robot state, task progress, and safety context to the operator. |
| D | Demonstration task is completed consistently within a defined time threshold, supported by integrated XR scene feedback and visual cues. |
| HD | Visual outputs support quantitative comparison between freeform and assisted modes, including clear representation of active safety regions, control state, or performance overlays. |

**Proposed:**

| Grade | Summary | Criteria |
|---|---|---|
| **P** | XR user interface for robtot data | • An XR user can observe pose markers and robot transforms information within a Unity XR UI element displayed in the scene. |
| **C** | XR object Interactions and scene visualisation | • Unity XR has visualisation of user interactable objects in the scene.<br>• Environment and UI elements are setup to support user interactions with task objects (e.g. sorting bins and object to sort).<br>• Object in Unity can be published to ROS2 and displayed in RVIZ. |
| **D** | XR and real-world environment digital twin | • Objects can be spawned into unity via perception stack allowing for ROS/perception to unity object interactions. |
| **HD** | High-fidelity scene with progress tracking and feedback | • More advanced UI elements for showing object sorting/task progress with Visual feedback on outcome (e.g. in progress or complete) |

| | | |
|---|---|---|
| | | • Scene is high fidelity with accurate models of interactable objects and environment features. |

# 9. Learning Goals

## 9.1 Team Learning Goals

- Apply systems engineering principles to design, integrate, and validate a multi-subsystem robotics solution.
- Develop and evaluate a full robotic sorting pipeline combining perception, planning, teleoperation, XR, and safety-aware operation in a real shared workspace.
- Strengthen interdisciplinary collaboration through clearly defined subsystem ownership and interface agreements.
- Improve technical communication through sprint reviews, documentation, and reflective critique.
- Develop resilience in debugging and integration of complex robotic systems.

## 9.2 Individual Learning Goals

- Gain practical experience integrating RGB-D perception data into real-time control and planning frameworks.
- Develop advanced understanding of autonomous pick-and-place using MoveIt 2 and collision-aware planning.
- Improve XR-based interface design for robotics applications, including gamification and operator UX.
- Strengthen hardware–software integration and safety-critical system thinking.
- Enhance professional capabilities in teamwork, reflection, and iterative design improvement.

# Appendix

**Appendix A**

Other potential names: HoloMotion, VisionLayer, Spatial Robotics, OmniSpace, Telessist, TeleMotion, HoloLens